

ODIHAM, United Kingdom

WMO: 037610

Lat: 51.2389N

Lon: 0.9450W

Elev: 123

StdP: 99.85

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.8	(c) -2.3	(d) -6.7	(e) 2.2	(f) -1.0	(g) -5.3	(h) 2.5	(i) 0.1	(j) 15.3	(k) 9.5	(l) 14.1	(m) 9.7	(n) 2.7	(o) 230	(p) 0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.0	(c) 26.8	(d) 18.1	(e) 24.7	(f) 17.3	(g) 22.9	(h) 16.5	(i) 19.1	(j) 24.7	(k) 18.2	(l) 22.8	(m) 17.4	(n) 21.4	(o) 4.2	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	12.5	20.6	16.5	11.9	19.7	15.7	11.3	19.0	54.8	24.9	51.8	22.9	49.2	21.4	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	10.8	9.6	-7.2	29.9	1.8	2.2	-8.5	31.5	-9.6	32.8	-10.6	34.0	-11.9	35.6
			-7.5	21.0	1.8	1.1	-8.8	21.8	-9.8	22.5	-10.8	23.1	-12.1	23.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	4.6	4.9	6.7	9.0	12.1	15.0	17.2	17.0	14.3	11.2	7.5	5.1
	DBStd	5.40	3.33	3.22	2.93	2.98	2.95	2.50	2.50	2.40	2.33	2.84	3.12	3.68
	HDD10.0	744	168	144	108	52	12	0	0	0	1	19	86	155
	HDD18.3	2927	425	376	362	279	195	103	52	57	121	221	326	410
	CDD10.0	899	1	1	4	22	76	152	224	216	131	58	10	3
	CDD18.3	40	0	0	0	0	1	4	18	15	2	0	0	0
	CDH23.3	340	0	0	0	1	9	50	153	113	13	1	0	0
	CDH26.7	61	0	0	0	0	0	7	28	24	1	0	0	0
(14) Wind	WSAvg	4.6	5.5	5.3	5.0	4.5	4.4	4.2	4.1	4.1	4.2	4.6	4.7	5.2
(15) Precipitation	PrecAvg	799	86	56	63	55	55	57	50	56	73	80	83	86
	PrecMax	1071	156	189	164	112	144	152	112	118	189	208	208	206
	PrecMin	557	0	0	2	2	6	6	10	8	7	8	23	15
	PrecStd	106	47	40	35	28	30	35	27	28	50	52	47	43
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	13.1	17.0	22.1	24.7	27.6	29.3	29.4	25.2	20.0	15.3	13.3
		MCWB	10.8	10.0	10.9	13.9	16.3	18.2	19.0	19.2	17.6	16.0	13.8	11.8
	2%	DB	11.4	11.5	14.3	18.5	22.4	24.6	27.1	26.4	22.3	17.6	14.1	12.5
		MCWB	10.2	9.6	9.7	12.2	15.3	17.1	18.0	18.4	16.6	14.6	12.7	11.2
	5%	DB	10.5	10.5	12.6	16.1	20.0	22.4	24.9	23.8	20.2	16.4	13.2	11.6
		MCWB	9.3	8.8	9.3	10.9	14.2	16.0	17.4	17.1	15.3	14.1	11.7	10.4
	10%	DB	9.6	9.5	11.2	14.2	17.6	20.5	22.8	21.8	18.7	15.5	12.2	10.6
		MCWB	8.4	8.0	8.6	9.9	12.9	15.1	16.7	16.3	14.7	13.6	10.8	9.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	11.1	11.8	14.7	17.7	19.7	20.4	20.5	18.7	16.7	14.3	12.3
		MCDB	11.9	12.4	14.4	20.4	22.1	26.2	27.1	27.4	22.7	19.3	14.9	12.8
	2%	WB	10.5	10.1	10.7	12.9	16.1	18.0	19.1	19.0	17.5	15.6	13.0	11.6
		MCDB	11.2	11.0	12.9	17.5	20.5	22.7	25.0	24.2	20.7	16.9	13.8	12.3
	5%	WB	9.5	9.2	10.1	11.6	14.8	16.7	18.1	18.0	16.5	14.7	12.0	10.6
		MCDB	10.3	10.2	11.9	14.7	18.9	20.9	23.3	22.0	19.0	15.9	13.0	11.6
	10%	WB	8.5	8.3	9.2	10.6	13.6	15.8	17.2	17.2	15.6	14.0	11.0	9.6
		MCDB	9.5	9.4	10.9	13.5	17.0	19.5	21.5	20.7	17.9	15.2	12.1	10.6
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	5.7	7.2	8.4	8.7	8.8	9.0	8.7	8.2	6.7	5.6	5.0
		MCDBR	5.6	6.2	9.4	11.5	12.4	12.4	12.7	12.1	10.5	7.1	5.6	4.7
		MCWBR	5.4	4.9	5.9	6.3	6.4	6.2	5.6	5.4	5.6	4.8	4.8	4.7
	5% WB	MCDBR	5.5	5.6	7.8	10.2	11.0	10.8	10.9	10.0	8.8	6.2	5.3	5.0
		MCWBR	5.5	5.0	5.6	6.0	6.2	6.0	5.5	5.1	5.5	4.8	5.0	5.1
(40) Clear-Sky Solar Irradiance	taub		0.326	0.341	0.377	0.398	0.402	0.400	0.394	0.386	0.370	0.354	0.328	0.315
	taud		2.373	2.381	2.277	2.234	2.265	2.297	2.320	2.361	2.399	2.427	2.418	2.358
	Ebn at Noon		685	781	813	838	853	856	855	843	815	755	679	631
	Edh at Noon		63	81	107	125	127	125	120	109	95	77	60	55
(44) All-Sky Solar Radiation	RadAvg		0.82	1.50	2.62	4.08	4.94	5.47	5.18	4.32	3.24	1.86	0.98	0.65
	RadStd		0.07	0.19	0.35	0.44	0.31	0.54	0.47	0.34	0.26	0.16	0.11	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (81 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page